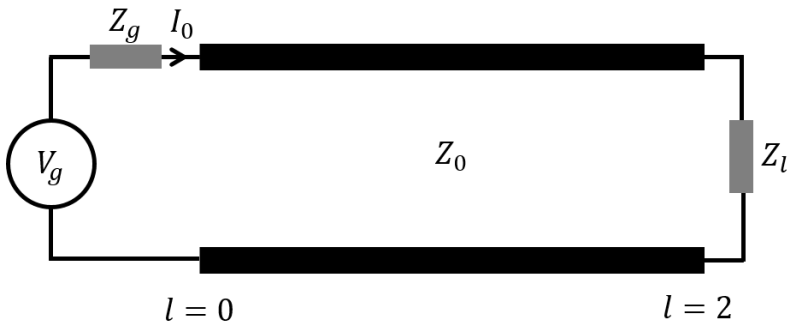
Winter term 2020/2021: Exercises "Radio Frequency Technology"

EXERCISE SESSION 2 (13.11.2020)

Problem 2.1: Transmission line with source and load

A transmission line of length $l = 2$ m is operating at $\omega = 10^6$ rad/s and has $\alpha = 8$ dB/m, $\beta = 1$ rad/m and characteristic impedance of $Z_0 = 60 + j40\ \Omega$ is terminated by a load of $Z_L = 20 + j50\ \Omega$. It is fed by a source of $V_g = 10\angle 0°$ V and $Z_g = 40\Omega$.



a) Calculate current I_0 on the driving side at $l = 0$.
b) Calculate current $I(l = 1)$ at midway distance of the transmission line.

Problem 2.2: Basics of Smith Chart

A transmission of characteristic impedance $Z_0 = 75\ \Omega$ is terminated by a load of $Z_L = 100 + j150\ \Omega$. Using Smith chart only calculate the following:

a) Load reflection coefficient Γ_L.
b) The corresponding standing wave ratio.
c) Load admittance Y_L.
d) Input impedance Z_{in} at a distance of 0.4λ from the load.
e) The transmission line is fed by a source and the length of the interconnecting transmission line is 0.6λ. Calculate location of voltage maximum and voltage minimum along the transmission line from the load to source
f) Finally, the input impedance seen by the source.

Problem 2.3: Stub matching

Antenna with input impedance of 40 +j30 Ω is to be matched to a 100 Ω lossless line with a shorted stub. Using Smith chart, determine

a) The required stub impedance
b) The distance of the stub from the antenna
c) The length of the stub
d) The standing wave ratio on the stub line, transmission line between stub and antenna and transmission line between generator and stub.